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Set 13

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City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements
2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.
3. Build a table to display the geographic data for each city. Label the table elements.
4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

Solution:

Code:

Import pandas as pd

Import matplotlib.pyplot as plt

Import folium

```
Df = pd.DataFrame({  
    'City': ['City A', 'City B', 'City C'],  
    'Population': [500000, 700000, 600000],  
    'Temperature': [75, 68, 80],  
    'Elevation': [1000, 800, 1200],  
    'Latitude': [13.08, 12.97, 11.01],  
    'Longitude': [80.27, 77.59, 76.96]
```

```

})

M = folium.Map(location=[12.5, 78.5], zoom_start=6)

For I in range(len(df)):

    Folium.Marker(

        [df.Latitude[i], df.Longitude[i]],

        Popup=df.City[i]

    ).add_to(m)

Display(m)

Plt.scatter(df['Temperature'], df['Population'])

For I in range(len(df)):

    Plt.text(df['Temperature'][i], df['Population'][i], df['City'][i])

Plt.xlabel('Average Temperature')

Plt.ylabel('Population')

Plt.title('Temperature vs Population')

Plt.grid()

Plt.show()

Display(df[['City', 'Population', 'Temperature', 'Elevation']])

```

Output:

